

13/07/24

Data Science Process

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1) Problem Definition

- Define Business → Define KPI Metrics
- Set objectives
- Identify stakeholders

2) Data Collection

- Relational DBs
 - No SQL
 - Web API's
 - Web Scraping
- Data Acquisition
Initial Storage

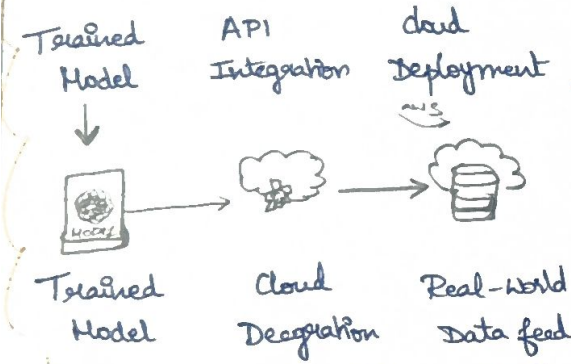
3) Data Preprocessing & cleaning

- Handle Missing values → Drop Impute
- Remove Duplicates
- Handle outliers
- Box plot

4) Exploratory Data Analysis (EDA)

- Univariate Analysis: Histogram
- Bivariate Analysis: Correlation Coefficient, Correlation Matrix
- Identify Trends
- Data Relationships

Deployment & Monitoring



Data Science Lifecycle

Model Evaluation

Confusion Matrix

A	1	0
1	5	0
0	0	1

	Precision	Recall
Accuracy	0.75	0.24
Precision	0.33	0.09
Recall	0.75	0.31
F1 Score		



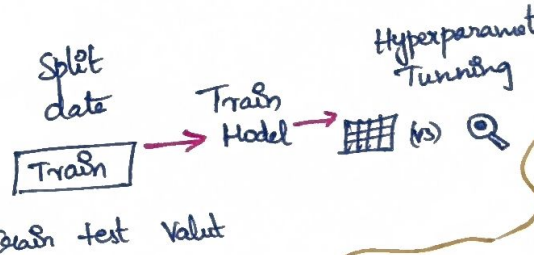
Cross-Validation



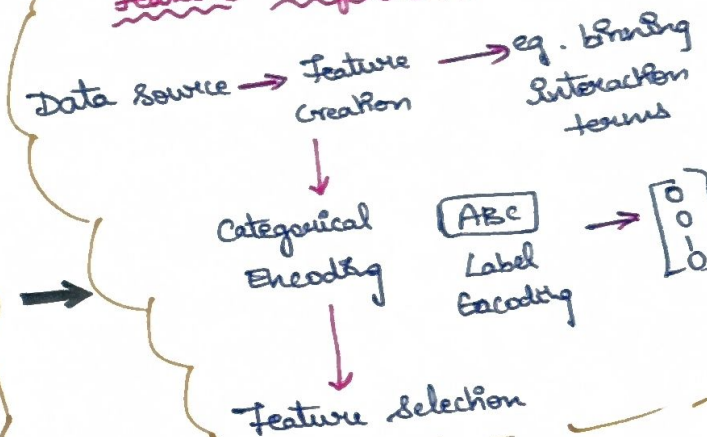
Model Building

Choose algorithm type

- Supervised: Linear Regression, Decision tree, SVM, K-Nearest
- Un-supervised: PCA



Feature Engineering



Performance Monitoring

Performance Monitoring



95%

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Fundamental of Data Science